

Text summarization: fundamentals

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di Torino**

Lecture goal

- The data deluge
- Data summarization
- Problem statement
- The Rouge score

The data deluge

- Online news
 - Newspapers and TV provide content on the web



- User Generated Content (Web & Mobile)
 - E.g., Facebook, Instagram, Yelp, TripAdvisor, Twitter, YouTube



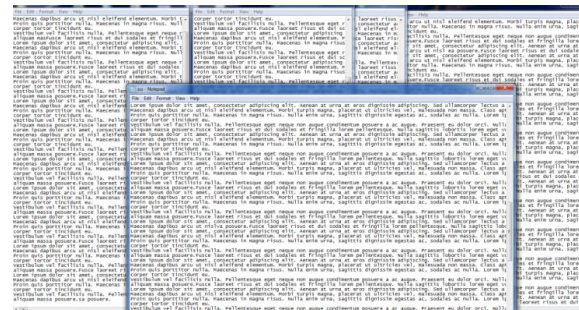
The data deluge

- E-learning platforms
 - Course materials & notes, homeworks & reviews



coursera

- Log files
 - Web server log files, machine syslog files



Data summarization

- Summary

“using few words to give the most important information about something”



- Data summarization pinpoints relevant information from large data collections

- large body of diverse techniques
- appropriate for different data types

Textual data summarization

- News

- Extract summaries of news articles ranging over the same topic and acquired from different sources



- E-commerce data

- Summarize the comments of the customers on the purchased products/services



Textual data summarization

- E-learning platforms

- Provide summaries of learning notes/teaching material to students in support of individual/collective learning activities



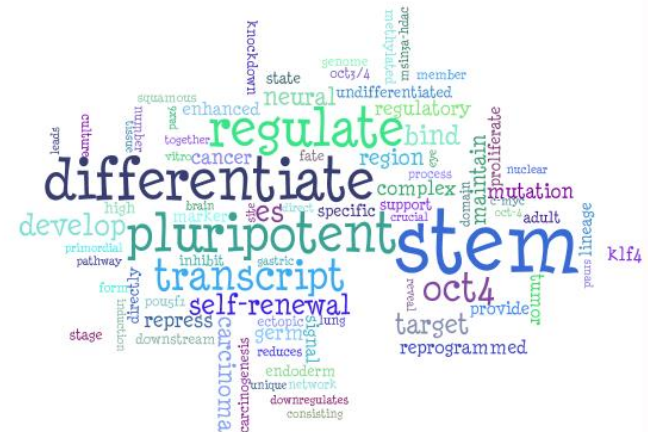
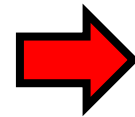
- Social networks

- Summarize the opinions expressed in online communities through the analysis of blogs and/or posts
- Topic, trend analysis



Why summarizing?

- Generate a big picture of the analyzed documents



Why summarizing?

- Quickly access key document information
 - disregarding redundant or marginally relevant information



- Guarantee content accessibility in case of limited bandwidth or reduced visualization quality



Text summarization

- Problem
 - Generation of a summary of a collection of textual documents
- Issues
 - Collection size
 - Single document vs multiple documents
 - Document characteristics and structure
 - E.g., semi-structured vs. unstructured
 - Language
 - Collection updates
 - Summary goal

Single-document summarization

- Problem
 - Generation a summary of a single document
- Examples
 - Abstracts of a scientific paper
 - Crib of a teaching document

Single-document summarization

- Advantages
 - Reduced computational complexity
 - Limited scalability issues
 - As long as the complexity of the analyzed document is fairly low
- Disadvantages
 - Need for exploration of the document structure
 - Homogeneous portions of text
 - Higher sensitivity to noise

Multi-document summarization

- Problem
 - Generation of a summary of a collection of documents
- Examples
 - Summary of the news articles published on different online newspapers and related to the same topic
 - Summary of all the online reviews of the same product

Multi-document summarization

- Advantages
 - Lower sensitivity to noise
 - Lower impact of the document structure
- Disadvantages
 - Higher computational complexity
 - Need for clustering homogenous documents
 - Presence of imbalances in the document distribution may bias the results
 - Scalability issues
 - Memory and time

Multi-lingual summarization

- Problem
 - Summarize collections of documents potentially written in different languages
 - Documents within the same collection are **all written in the same language**
- Challenges
 - Need for solutions portable to different languages
 - No need for combining models related to different languages

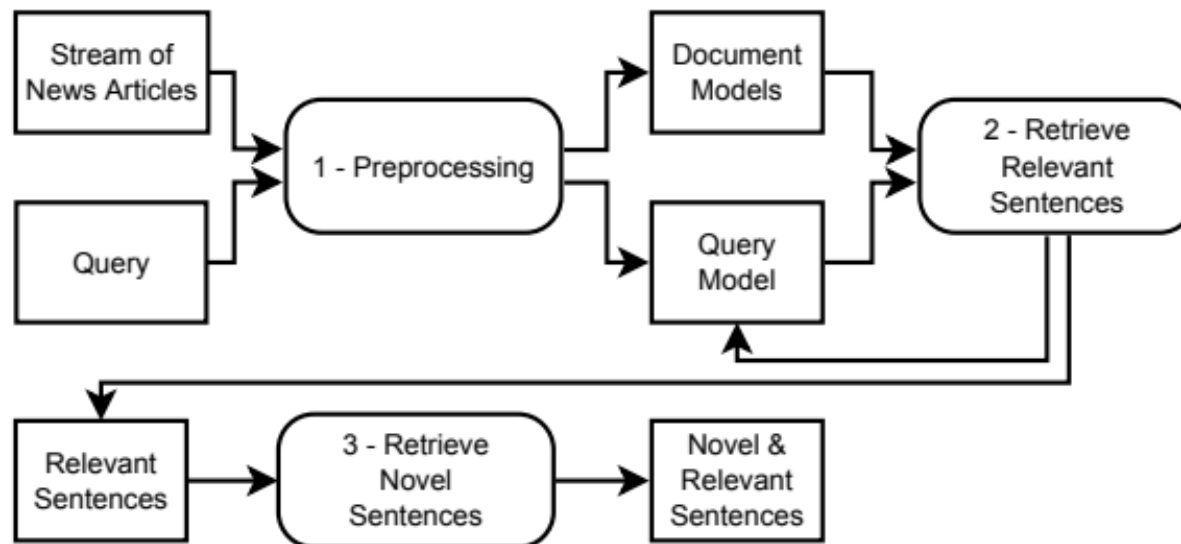
Cross-lingual summarization

- Problem
 - Summarize collections of documents potentially written in different languages
 - Documents within the same collection **can be written in different languages**
- Challenges
 - Need for solutions portable to different languages
 - Need for combining models related to different languages

Temporal summarization

● Problem

- Summarize a timestamped sequence of textual documents
 - E.g., a news stream
- Identify salient concepts with minimal latency and redundancy



T. Schubotz and R. Krestel, "Online Temporal Summarization of News Events,"
2015 IEEE/WIC/ACM WI-IAT, Singapore, 2015, pp. 409-412. doi: 10.1109/WI-IAT.2015.159

Multimodal summarization

- Problem statement
 - Generate textual summaries of data generated from multimodal sources
 - Adapt textual summaries to specific contexts, users, or data platform
 - E.g., Use metadata extracted from videos

Transcript

today we are going to show you how to make spanish omelet . i 'm going to dice a little bit of peppers here . i 'm not going to use a lot , i 'm going to use very very little . a little bit more then this maybe . you can use red peppers if you like to get a little bit color in your omelet . some people do and some people do n't t is the way they make there spanish omelets that is what she says . i loved it , it actually tasted really good . you are going to take the onion also and dice it really small . you do n't want big chunks of onion in there cause it is just pops out of the omelet . so we are going to dice the up also very very small . so we have small pieces of onions and peppers ready to go .



Summary

how to cut peppers to make a spanish omelette; get expert tips and advice on making cuban breakfast recipes in this free cooking video .

Multimodal Abstractive Summarization for How2 Videos.
Shruti Palaskar, Jindrich Libovicky, Spandana Gella, Florian Metze. <https://arxiv.org/abs/1906.07901>

Multimodal summarization

- Multimodality can be either at the input or at the output layers
- When the modality changes from input to output: **Cross-modal summarization**

Transcript

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Generic vs. query-relevant summarization

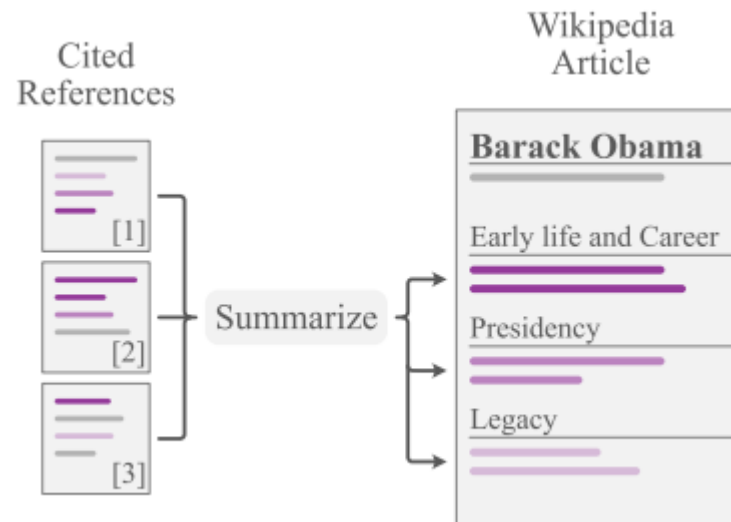
- Generic document summarization
 - The summary should reflect the major content of the documents without any additional information
 - Model generation is solely based on the analyzed documents
- Query-relevant document summarization
 - The summary focuses on the information expressed in the given queries
 - the summaries must be biased to the given queries
 - Model generation should take query content into account

Query-relevant summarization vs. IR

- Summaries have further constraints beyond pertinence
 - E.g., Minimal redundancy, minimum latency, etc.
- Summaries can be abstractive
- The ordering of the summary content is not always relevant
 - E.g., Generate a summary consisting of 665 bytes

Aspect-based summarization

- Generic document summarization
 - Allow readers to fulfill focused information needs more easily and quickly
- Can be either domain-specific or open-domain
 - Whether they are applicable to various domains or not



WikiAsp: A Dataset for Multi-domain Aspect-based Summarization. Hiroaki Hayashi, Prashant Budania, Peng Wang, Chris Ackerson, Raj Neervannan, Graham Neubig. ACL 2020

Aspect-based summarization

- Aspects can be derived from either section titles or further document annotations
 - E.g., Wikipedia sections

Title: Barack Obama

Aspect: Early life and Career

Obama was born on August 4, 1961, at Kapiolani Medical Center for Women and Children in Honolulu, Hawaii.

...

Aspect: Presidency

The inauguration of Barack Obama as the 44th President took place on January 20, 2009. In his first few days in office, Obama issued ...

Aspect: Legacy

Obama's most significant legacy is generally considered to be the Patient Protection and Affordable Care Act (PPACA), ...

Example of aspect-based summarization

- Generate Wikipedia pages automatically by using cited references as document source for summarization
 - E.g., books, websites, etc.
- Let $\{R_1, R_2, \dots, R_M\}$ be a collection of M cited references for an article $S = \{s_1, s_2, \dots, s_N\}$ of N sections
- Problem: learn a model $f : R \rightarrow S$
 1. identify and gather information from cited references
 2. generate a section-by-section summary, where each section contains the appropriate type of information: section title and one or more paragraphs

WikiAsp: A Dataset for Multi-domain Aspect-based Summarization. Hiroaki Hayashi, Prashant Budania, Peng Wang, Chris Ackerson, Raj Neervannan, Graham Neubig. ACL 2020

Incremental summarization

- Problem

- Generate a summary from the input document collection
- As soon as the collection changes the summary content should be updated
 - Incremental models are needed to avoid re-calculating the model from scratch

- Challenges

- Most data mining techniques are non-incremental
- The relevance of the old document content should be updated according to the new content

Evaluation methods

- Evaluate the quality of the output summaries is crucial for assessing the effectiveness of automated summarization systems
- Quantitative evaluation
 - Similarity with the ground truth
- Qualitative evaluation
 - Readability, pertinence, smoothness, etc.

Evaluation methods

- Intrinsic evaluators
 - Assess the quality of a summary against a reference summary or based on the feedbacks provided by a domain expert who manually evaluated the output summaries
- Extrinsic evaluators
 - Measure the effectiveness of a summarization method in accomplishing a given task

The Rouge score

- Recall-Oriented Understudy for Gisting Evaluation
- Intrinsic evaluation metric
 - Based on syntax
- It allows analysts to compare an automatically generated summary with a set of reference summaries
 - Typically human-generated
- It measures the unit overlaps between the text of the reference summary, i.e., the ground truth, and text contained in the automatically generated summary

N	ROUGE metrics		
	<i>Recall</i>	<i>Precision</i>	<i>F-measure</i>
ROUGE-1	0.5713	0.5658	0.5476
ROUGE-2	0.4710	0.4597	0.4465

Rouge-N

- N-gram co-occurrences between reference and candidate summaries
- High order ROUGE-N with n-gram length greater than 1 estimates the fluency of summaries
- Rouge-N recall:

$$\begin{aligned} & \text{ROUGE-N} \\ &= \frac{\sum_{S \in \{\text{ReferenceSummaries}\}} \sum_{gram_n \in S} \text{Count}_{\text{match}}(gram_n)}{\sum_{S \in \{\text{ReferenceSummaries}\}} \sum_{gram_n \in S} \text{Count}(gram_n)} \end{aligned}$$

Rouge-N measures

- Standard Precision, Recall, and F1 score measures
- Rouge-N recall:

$$\text{ROUGE-N} = \frac{\sum_{S \in \{\text{ReferenceSummaries}\}} \sum_{gram_n \in S} \text{Count}_{\text{match}}(gram_n)}{\sum_{S \in \{\text{ReferenceSummaries}\}} \sum_{gram_n \in S} \text{Count}(gram_n)}$$

- Considering Rouge-N Recall is advisable while evaluating fixed-size summaries
 - Recommendation: use Rouge-N F1 score otherwise!

Rouge-N

- Rouge evaluation ignores semantic similarity
 - Example
 - reference text: *police killed the gunman*
 - Summary 1: police kill the gunman
 - Summary 2: the gunman kill the police
 - The two summaries above have the same N-gram co-occurrence!

Rouge-L

- Longest Common Subsequence
 - Given two sequences X and Y, the longest common subsequence of X and Y is a common subsequence with maximum length
 - Intuition: the longer the match, the more similar the summaries
- Used for comparing text translations as well
- Example
 - reference text: *police killed the gunman*
 - Summary 1: police kill the gunman -> LCS overlap 3/4
 - Summary 2: the gunman kill the police -> LCS overlap 2/4
 - Summary 1 is better than Summary 2 ($3/4 > 2/4$)
- The two summaries above have the same N-gram co-occurrence!

Rouge-W

- Weighted Longest Common Subsequence
- Favors strings with consecutive matchings
- Can be computed efficiently using dynamic programming
- Example
 - Reference: [A B C D E F G]
 - S1: [A B C D H I K]
 - S2: [A H B K C I D]
 - Rouge-L(S1) = Rouge-L(S2)
 - Rouge-W(S1) > Rouge-W(S2)

Rouge-S

- Skip-Bigram
- Any pair of words in their sentence order, allowing for arbitrary gaps
- Intuition
 - Consider long distance dependency
 - Allow gaps in matches as in LCS but count all in-sequence pairs rather than longest subsequences only

Rouge-S example

- Example

- reference text: *police killed the gunman*
- Summary 1: police kill the gunman
- Summary 2: the gunman kill the police
- Summary 3: the gunman police killed

- Results

- Rouge-N: Summary 4 > Summary 2 = Summary 3
- Rouge-L: Summary 2 > Summary 3 = Summary 4
- Rouge-S: Summary 2 > Summary 4 > Summary 3
 - Summary 2: 3/6 (“police the”, “police gunman”, “the gunman”)
 - Summary 3: 1/6 (“the gunman”)
 - Summary 4: 2/6 (“the gunman”, “police killed”)

Rouge-SU

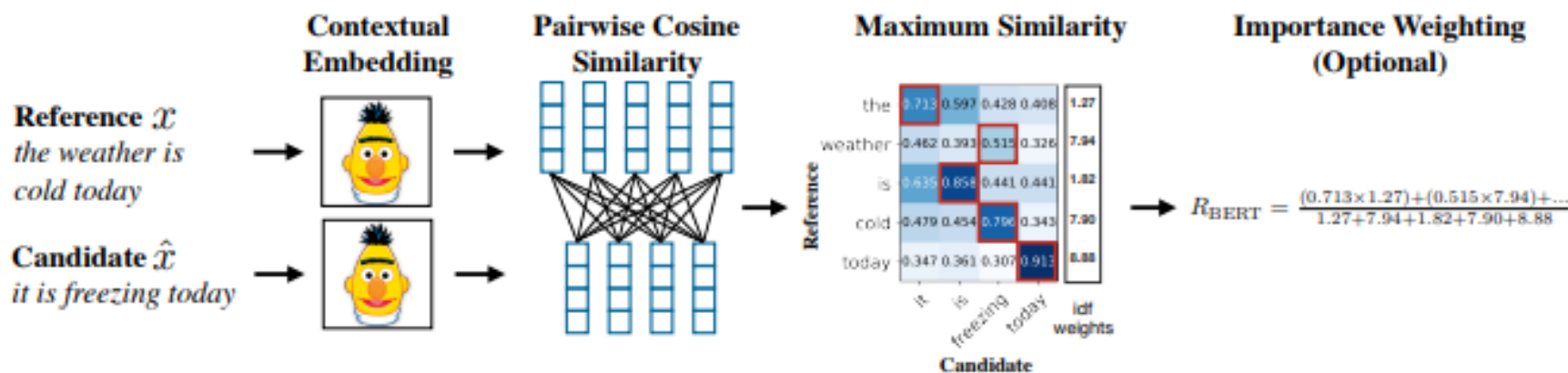
- Skip-Bigram plus unigram-based co-occurrence statistics
- Similar to Rouge-S
- It counts **both** unigrams and bigrams

JRouge

- Java-based Rouge extension designed for non-European languages
- Improvements
 - Uses unicode regular expressions to match characters and numerals of any Unicode text
 - Uses the Stanford NLP toolkit to split the sentences
 - Provides consistent results in different Operating Systems
 - Calculates scores more precisely than the original ROUGE since it does not truncate and round the results to the fifth digit after the decimal point. Instead, JRouge uses the whole extent of the Double primitive type of Java to provide precise scores

BERT score

- Automatic evaluation metric for text generation
- Correlates human judgments using BERT
- Suitable for both abstractive and extractive summarization



BERTScore: Evaluating Text Generation with BERT. Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, Yoav Artzi. ICLR 2019.

BERT score

- Precision, Recall, and F1 Score metrics

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \quad P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \quad F_{\text{BERT}} = 2 \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

where the similarity between the reference and candidate tokens is computed using the cosine similarity $\frac{\mathbf{x}_i^\top \hat{\mathbf{x}}_j}{\|\mathbf{x}_i\| \|\hat{\mathbf{x}}_j\|}$

BERTScore: Evaluating Text Generation with BERT. Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, Yoav Artzi. ICLR 2019.

LLM-As-an-Expert

- Use LLMs to assess summary quality
- Typically, using comparative analysis
- A/B Testing
 - randomized experimentation process where multiple summary versions of various documents are evaluated by an LLM “as a judge” to determine which version is the best one

Mean Reciprocal Rank

- Extrinsic evaluation metric, established in Information Retrieval
- Input:
 - a query, a ranked list of retrieved object, and the expected ranked list of objects
- Output
 - inverse of the rank of each of the retrieved objects

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^Q \frac{\text{Relevance_Label_Value}}{\text{rank}_i}$$

Mean Reciprocal Rank

- Can be tailored to extractive text summarization
- Example
 - Object: sentence.
 - Summary: ranked list of sentences.
 - Golden summary: expected ranked list of sentences.

Benchmark datasets (small)

- **Document Understanding Conference 2001**
 - single- and multi-document summarization task
 - 60 collections of documents which have been partitioned into training and test sets
- **Document Understanding Conference 2004**
 - Multi-document summarization
 - variant of the DUC 2001 dataset
- **Document Understanding Conference 2007**
 - Variant of the DUC'04 collection
 - update summarization task
- **MultiLing Pilot 2011**
 - Multi-lingual summarization
 - A set of WikiNews documents translated in Arabic, Czech, English, French, Greek, Hebrew and Hindi by native speakers
 - Each article belongs to one of the 10 topics under consideration and for each topic at least 3 reference summaries are provided
- **Text REtrieval Conference 2013**
 - Temporal summarization task
 - It consists of a collection of news articles, which have been chronologically ordered to simulate a stream of news articles ranging over a set of events
 - Tasks: filtering and summarization or summarization only

Benchmark datasets (large)

- **CNN/DailyMail dataset**
 - Multi-document news summarization
- **Big Patent dataset**
 - Single-document patent summarization
- **TLS-Covid19**
 - Sars-Cov-2 news timeline summarization
- **ScisummNet**
 - Scientific paper summarization based on cited text spans analysis

Additional reading on ROUGE



- ROUGE: A Package for Automatic Evaluation of Summaries. Ming-Wei Chang Kenton Lee Kristina Toutanova. Proceedings of the ACL 2004.
- Download and read the paper: <https://aclanthology.org/W04-1013.pdf>

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- Affiliation

- The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
 - <https://dbdmg.polito.it>
 - <https://smartdata.polito.it>

Thank you!